

Project Esky: Enabling High Fidelity Extended Reality on an Open Source Platform

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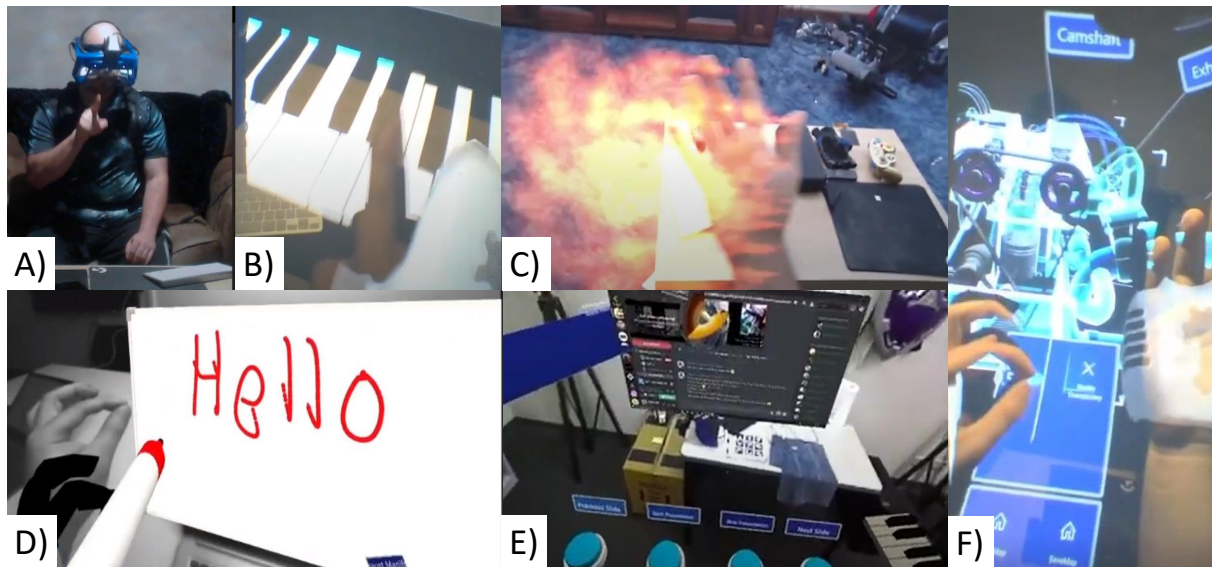


Figure 1: We present several demonstrations made with Project Esky, A complete open source extended reality (XR) framework capable of high fidelity experiences. A) A user wearing an optical see-through HMD, and the connected backpack PC. B) A photo from the user's view as they interact with a virtual piano. C) A re-imagining of BurnAR, a system which emits virtual fire from a user's hands. D) A white board and model drawing application, allowing a user to paint on real and virtual objects. E) A virtual desktop control system designed specifically for online live stream demonstrations and presentations. F) An educational example with a virtual annotated car engine.

ABSTRACT

This demonstration showcases the improvements made to Project Esky[2], an open source Extended Reality (XR) software framework capable of high fidelity natural hand-interactions with virtual content. In combination with showing our temporal re-projection rendering technique in realtime, we also showcase several live desktop demonstrations, such as a virtually controlled powerpoint presentation, an e-learning example with an animated car engine, a fire simulation and a model painting application. It is our hope that the demonstrations presented will inspire others to work with

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Project Esky to build their own experiences, and will be a stepping stone towards bringing high-fidelity XR content to researchers and hobbyists.

CCS CONCEPTS

• **Software and its engineering** Development frameworks and environments; • **Human-centered computing** Interactive systems and tools; • **Human-centered computing** Mixed / augmented reality;

KEYWORDS

Augmented Reality, Mixed Reality, Extended Reality, Development tools, Software Frameworks

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1 SYSTEM DESIGN AND IMPLEMENTATION

This section describes the platform which we deploy Project Esky for the purposes of the desktop demonstrations. At the high level, Project Esky is designed to be a modular system, with components that are easily substituted, allowing for better hardware or cheaper solutions as per user specific requirements. We built our variation of Project Esky using the following hardware:

- PC: HP Z VR G2 Backpack PC. 2.6 GHz Intel Core i7-8850H, 32 GB RAM, GTX 2080
- Display: Project Northstar [4]
- Hand Tracking: Ultraleap Leap motion rigel
- 6DOF tracking: Intel Realsense T261

For software, we utilize Project Esky's low level API c++ plugins to handle undistortion rendering, tracking, and temporal reprojection, which are consumed by the Unity Game Engine via. several High level C# APIs. The high level API plugins also integrates Microsoft's Mixed Reality Toolkit (MRTK) with the LeapMotion tracker to provide hand-gesture interactions with virtual content.

2 DEMONSTRATING THE CAPABILITIES OF PROJECT ESKY

As shown in the accompanying video, our open source software framework is capable of high fidelity XR. In addition to showing the new temporal re-projection feature built into Project Esky by running the system at a much lower frame rates (15-30Hz) than the display is capable of (120Hz), we showcase several desktop demonstrations that have been quickly built using the combined software alongside the default demonstrations packaged with the integrated Mixed Reality Toolkit.

Virtual Piano: This demonstration focuses on the finite finger tracking afforded by our platform. The user is presented a virtual piano that has been connected to a realtime-generation of an synthesized organ using Virtual Studio Technology (VST). The piano key velocity is directly translated to the VST, producing a realistic sound effect.

E-learning Engine: This demonstration is an e-learning example, presenting the users with a virtual rendering of a toyota 4AGE engine with attached tooltip annotations, labelling the parts. Users are able to manipulate the virtual car engine, with full animations of the pistons, crankshaft, and all extra components. The hand menu allows users to turn the engine transparent (to show the internal engine components) and adjust the speed the engine rotates.

Burning Hands: This demonstration is inspired by BurnAR [5], and a more recent implementation by Eckhoff et al [3]. As we are able to map a hand model to the user's tracked hand, we convert it to a signed distance field, and generate a volumetric rendering of fire particles from the SDF representation of the user's hand. The resulting effect is a realistic fire rendering. This demonstration showcases the high quality afforded by the backpack PC in an interactive setting.

Whiteboard and Model Painting: In this demonstration, a user is able to use virtual markers, and their fingers, to paint on virtual or real objects. The latter is performed by rendering a phantom model plane of a flat surface. We then calculate the intersection point between the virtual paint marker and the virtual plane/model, writing to the texture according to the calculated UV map at the intersected point.

Virtual Desktop, and Powerpoint Presentation: As part of the demonstration suite shown at Interactive Surfaces and Spaces 2020 [1], we will present the technical components and implementations of Project Esky as a powerpoint presentation controlled entirely by the virtual interface, rendering the desktop and presentation preview windows in the virtual space.

Viewer Participation: Since the conference will be held virtually, we intend to showcase the applications built with Project Esky as a live online streamed demonstration. Initially viewers will be shown a powerpoint slide presentation, controlled by the virtual interface. Afterwards, while the live actor demonstrates the system, viewers can preview the demo with through-the-lens and video see-through previews of the action occurring. Participants are able to pseudo-guide the live actor, and request the actor interact with which demonstration they please. In particular, we intend to allow participants to dictate what is drawn in the whiteboard or model painting applications, later sharing the resulting drawings.

3 CONCLUSION AND FUTURE WORK

In these demonstrations, we showcase the improvements made to Project Esky, and some of Project Esky's capabilities via a live demonstration of our temporal re-projection technique and several high quality desktop XR experiences. It is our hope that attendees who view our demonstrations are inspired by the designs of our XR experiences. This will hopefully encourage attendees to build their own experiences with Project Esky, bringing high fidelity XR to the masses.

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A ONLINE RESOURCES

Supplemental Video (Online): https://youtu.be/EyobMiSPp_I